

Chapter 139 — Surgical Management of Upper Urinary Tract Urothelial Tumours

DIAGNOSIS

Diagnostic Endoscopic Evaluation and Biopsy

- **Suspicion usually arises on imaging:** a soft-tissue mass involving the collecting system of the renal pelvis or ureter raises the possibility of upper tract urothelial carcinoma (UTUC).
- **Endoscopy with biopsy and selective cytology is the most accurate means of diagnosis.** Obvious cases can be identified by good-quality computed tomography (CT) or magnetic resonance imaging (MRI) urography combined with cystoscopy and positive urinary cytology.
- **Diagnostic accuracy rises from about 75% with excretory or retrograde urography alone to 85–90% when ureteroscopy is added.** Endoscopic assessment with biopsy is therefore the **gold standard for diagnosis, as an adjunct to contrast-enhanced cross-sectional imaging.**
- **Diagnostic endoscopy carries oncologic risk:** bladder tumour recurrence, pyelovenous and pyelolymphatic migration of cancer cells, and other routes of dissemination. Special care is required (see Ureteroscopy below).
- **Diagnostic versus therapeutic endoscopy:** diagnostic endoscopy is a brief, low-impact procedure confined to visual characterisation of involved sites and tissue sampling (biopsy and cytology); therapeutic endoscopy has treatment intent. The two may be combined in one sitting for **small, easily accessible tumours.**
- **Percutaneous antegrade pyelo-ureteroscopy** is used when retrograde ureteroscopy is not feasible, for example after urinary diversion.
- **Image-guided percutaneous needle sampling** (CT- or ultrasound-guided) is the alternative when tissue is needed but endoscopy is considered less safe.
- **Endoscopy also shows areas not seen on imaging** and informs treatment planning. The American Urological Association (AUA) / Society of Urologic Oncology (SUO) guideline on non-metastatic UTUC derives its risk groups from endoscopic and imaging features, which makes a **standardised diagnostic assessment** essential.
- **Preoperative ureteroscopy has clear value when the diagnosis is uncertain or kidney-sparing management is being considered.**
- **Ureteroscopic biopsies are small and shallow, so their correlation with pathological stage is weak.** Stage is best estimated by combining radiographic findings, the endoscopic appearance of the tumour, biopsy grade and cytology.

Concept — *Because the biopsy cannot stage the tumour, UTUC is staged by inference: grade, appearance, imaging and cytology together stand in for depth of invasion, and every later risk-group decision rests on that composite.*

TREATMENT ALGORITHM

- **Radical nephroureterectomy (RNU) with bladder cuff resection is the standard therapy,** provided the contralateral kidney is normal and there is no surgical contraindication.
- **Risk-adapted strategies** from the AUA/SUO and the European Association of Urology (EAU) classify patients by tumour features to refine surgery, including kidney-sparing management (Box 139.1).

- **Low-risk disease → consider kidney-sparing treatment irrespective of the contralateral kidney,** preserving renal function without compromising oncologic outcome.
- **Kidney-sparing may also be considered in selected higher-risk (imperative) situations:** (1) serious renal insufficiency, (2) solitary kidney, (3) bilateral tumours, (4) poor surgical candidate. The treatment must not significantly compromise oncologic outcome; cancer-control risk is weighed against renal preservation through shared decision-making.
- **Lower-ureteric disease, particularly with unfavourable features → segmental resection rather than endoscopic management** as definitive treatment.

BOX 139.1 RISK-ADAPTED PROTOCOL FOR KIDNEY-SPARING MANAGEMENT

Feature	Low risk — favourable	Low risk — unfavourable	High risk — favourable	High risk — unfavourable
Grade	Low	Low	High	High
Cytology	Negative	No high-grade cytology	Any	High-grade
Radiographic invasion	None	None	None	Invasion
Obstruction	None	Obstruction	None	Obstruction
Nodes on imaging	Normal	Normal	Normal	Suspicious
Focality	Unifocal	Multifocal	Unifocal	Multifocal
Architecture	Papillary	Papillary	Papillary	Sessile or flat
Lower tract involvement*	No	Yes	No	Yes
Ablative treatment	Preferred	May be offered	Rare, select cases	Palliation only
Systemic therapy	Not recommended	Not recommended	Neoadjuvant or adjuvant	Neoadjuvant or adjuvant

*Prior or concomitant history of lower tract involvement. Modified from Coleman JA, Clark PE, Bixler BR, et al.: AUA/SUO guideline on non-metastatic UTUC, J Urol 209:1071–1081, 2023.

- **Grade makes the primary division;** substratification then turns on obstruction, multifocality and lower-tract involvement in low-grade disease, and on invasion, obstruction, suspicious nodes, multifocality and sessile or flat architecture in high-grade disease.
- **Ablation follows a gradient:** preferred → may be offered → rare, select cases → palliation only. Systemic therapy enters only when grade is high.
- **Trap: low grade does not automatically mean ablation** — obstruction or multifocality moves a low-grade tumour into the unfavourable group, where RNU retains a role.

Algorithm for localised UTUC (Fig. 139.16): radiographic suspicion → contrast-enhanced CT or MRI urogram → endoscopic examination with biopsy and cytology → risk-group assessment and shared decision → either **nephroureterectomy ± neoadjuvant chemotherapy (NAC)** for high risk with adequate glomerular filtration rate (GFR), or **conservative management** for low risk or poor GFR.

Concept — *The algorithm balances two axes, tumour risk and renal reserve: high risk with adequate reserve goes to extirpation, low risk goes to preservation, and the imperative indications are exactly where the two axes conflict.*

SURGICAL MANAGEMENT

- **Options mirror those for lower tract (bladder and urethral) cancer:** extirpative, endoscopic, topical, and systemic chemotherapy and immunotherapy (neoadjuvant and adjuvant).
- **The evidence base was historically thin** because UTUC is uncommon and heterogeneous; landmark trials and well-curated consortium data now define best practice, and better endoscopic optics and instruments allow less invasive diagnosis and treatment.
- **The least invasive treatment is preferred only if it does not compromise oncologic control.**
- **UTUC is unforgiving of surgical indiscretion:** it seeds and implants after tumour spillage or extravasation and is not amenable to salvage by other modalities, so every approach must avoid spillage.

Radical Nephroureterectomy

Indications

- **RNU with bladder cuff excision and lymph node dissection is recommended for non-metastatic, high-risk tumours of the renal pelvis and proximal ureter.**
- **High-risk disease:** high-grade cancer on biopsy or cytology, or clinical features of invasive or locally advanced cancer such as radiographic obstruction or parenchymal invasion (Box 139.1).
- **RNU also retains a role in low-risk, low-grade cancers with unfavourable features** (obstruction, multifocality, large volume) **or when endoscopic management is unsuccessful or infeasible.**

Techniques

Open radical nephroureterectomy

- **Multiquadrant access** is required to the kidney, perinephric fat, ipsilateral adrenal, retroperitoneal nodes, ureter and bladder, through a single incision or separate incisions.
- **Positioning:** supine or modified flank. The genitalia are included in the field so that the bladder catheter can be placed and accessed intraoperatively. After prior pelvic surgery, perineal access may be needed to manipulate the vagina or rectum for identification.
- **Midline incision gives the best exposure of retroperitoneal nodes and bladder but limits the upper pole,** particularly on the left and in obese patients.
- **Flank, subcostal, modified Makuuchi or thoracoabdominal incisions** can be extended toward the pelvis; **unless a midline incision is used, a second Gibson, lower midline or Pfannenstiel incision is needed for the bladder cuff** (Fig. 139.1).
- **Sequence — kidney first, cuff last,** so the collecting system stays intact until the end: incise white line of Toldt → mobilise colon to expose Gerota's fascia → **clip or ligate the ureter early** (limits migration of tumour cells or fragments to the bladder during renal dissection) → **control the hilum before excessive manipulation** → secure and divide renal artery and vein (suture ligature, ties, clips, or endovascular stapler) → mobilise the entire kidney **outside Gerota's fascia**, dividing hepatorenal attachments on the right and the splenorenal ligament on the left.
- **Side-specific hazards:** right — duodenum reflected medially to expose the hilum; left — pancreatic tail, spleen and fourth (ascending) part of the duodenum.
- **Routine adrenalectomy is unnecessary;** it does not aid oncologic control unless involvement is suspected on imaging or at operation.

Management of the distal ureter and bladder cuff

- **Complete removal of the distal ureter and bladder cuff is oncologically superior to incomplete resection.** Surveillance of a residual stump is clinically impractical and local recurrence is high, so **the entire distal ureter, including the intramural portion and the ureteric orifice, must be removed.**

- **A retained intramural stump carries local recurrence in 30–75%.** Non-definitive (incomplete) distal management carries higher local and distant recurrence and inferior disease-specific and overall survival.
- **After prior urinary diversion into bowel, the ureteroenteric anastomosis is excised with a bowel cuff;** recurrence is substantial when it is left.
- **The kidney may stay in continuity with the distal ureter,** or, because its bulk hinders manipulation, the ureter may be divided in a controlled manner between ties, clips or stapler at a grossly tumour-free level. **Do not seal the ureter with a thermal vessel-sealing device alone** — durable sealing of the ureter is off-label and unestablished.
- **No single technique is superior,** but techniques that are endoscopic-assisted or risk spillage outside the collecting system should be avoided:

DISTAL URETER TECHNIQUES THAT HAVE FALLEN OUT OF FAVOUR

Technique	Reason abandoned
Transvesical "lasso" procedure	Spillage risk; higher recurrence
Cystoscopic transmural resection ("pluck")	Spillage outside the collecting system; higher recurrence
Intussusception endoluminal "stripping"	Higher recurrence
Laparoscopic extravesical "staple"	Positive margins and staple-line recurrence — urothelial mucosa trapped in the staple line

- **General principles of distal ureterectomy:** excise the orifice with a small margin of surrounding urothelium → avoid or contain urinary spillage outside the collecting system → obtain a watertight bladder closure.

Transvesical and extravesical ureterectomy

	Extravesical (Fig. 139.2)	Transvesical (Fig. 139.3, combined)
Usual setting	Minimally invasive (laparoscopic, robotic)	Open , often via separate Gibson or Pfannenstiel incision
Access	Ureter followed distally to the intramural segment (landmark: passes under the superior vesical artery)	Anterior cystotomy; intravesical dissection of the ureter
Cuff	En bloc cuff, often by clamping the bladder wall (open)	Traditional 1-cm mucosal cuff ; wider if gross tumour protrudes; en bloc partial cystectomy if invasive intramural tumour suspected
Hazard	Uncontrolled entry into the collecting system while dissecting the tunnel; contralateral orifice	Contralateral orifice

- **Either is acceptable provided the whole ureter, including the intramural portion and the mucosa of the orifice, is removed with visual confirmation of complete resection.**
- **Extravesical sequence:** mobilise the ureter distally to the intramural segment → tie the ureter where it enters the detrusor tunnel (retraction and spillage control) → gentle traction with partial bladder filling through the urethral catheter → divide the lateral pedicle or its branches (obliterated umbilical, superior, middle and inferior vesical arteries) to expose the entire intramural ureter → flush and empty the bladder through the catheter → excise the cuff en bloc, avoiding the contralateral orifice. Traction everts the orifice outside the bladder (Fig. 139.2).

- **Robotic or laparoscopic cases:** modified flank position; enter the bladder at a supralateral site relative to the ureteric insertion so that it drains completely without spillage; a purse-string suture preplaced around the tunnel helps begin closure.
- **Closure:** two layers, interrupted or running absorbable — first mucosa, then detrusor and adventitia. Foley catheter to gravity for several days; suction drain in the perivesical space optional.

Concept — *Every distal-ureter decision follows from one fact: tumour cells in urine implant wherever urine goes, so the ureter is closed early, the orifice is removed whole, and the bladder is opened only when it is empty and the ureter is tied.*

Laparoscopic and Robotic-Assisted Radical Nephroureterectomy

Indications

- **Indications are the same as for open RNU.** Possible exceptions: large bulky tumours involving adjacent structures, and cases in which extended lymph node dissection is considered after chemotherapy.
- **Approaches:** transperitoneal, retroperitoneal, hand-assisted and robotic-assisted.
- **Retrospective non-randomised series show oncologic survival generally equivalent to open surgery in experienced centres;** less invasive procedures are better tolerated with fewer major complications, but **may carry more bladder recurrence and are less favoured for advanced (T3/T4) disease.**
- **Two components:** nephrectomy with proximal ureterectomy, then excision of the distal ureter with **intact specimen extraction.** Tissue handling must avoid spillage.
- **The extraction incision is placed to serve both extraction and distal ureteric dissection.** Because an extraction incision is needed anyway, techniques designed to avoid a second incision lose their utility when the distal ureter is managed open.

Technique

Transperitoneal laparoscopic nephroureterectomy — kidney down to mid-ureter

- **Positioning:** kidney and proximal ureter before the cuff; the patient is placed supine with the ipsilateral hip and shoulder rotated about 20–30° (the illustrated set-up, Fig. 139.4, uses a modified lateral decubitus with the flank rotated up 15°), secured at chest, waist and lower limb so that **table rotation** moves the patient between flank (nephrectomy) and modified supine (open distal ureterectomy). An alternative is slight outward rotation at the waist to flatten the lower abdomen.
- Genitalia and ipsilateral flank prepared and draped; **Foley catheter placed before insufflation.**
- **Ports:** three or four (Fig. 139.5). Camera at the umbilicus throughout; a 30° lens helps. Upper midline and lateral ports for the kidney and proximal half of the ureter; lower midline and lateral ports for the distal ureter; a 3- or 5-mm subxiphoid port for liver retraction on the right. **In larger abdominal girth, shift the ports laterally** (Fig. 139.6). A hand port, if used, is sited to serve the distal ureter and open cuff.
- **Choice of extraction incision** (Fig. 139.5): a low abdominal (midline or Pfannenstiel) incision if laparoscopic dissection has reached below the iliac vessels; a Gibson-type incision if the more proximal ureter needs exposure.
- **Colon:** incise along the line of Toldt from the iliac vessels to the hepatic flexure (right) or splenic flexure (left); mobilise the colon medially by releasing the renocolic ligaments while **leaving the lateral attachments of Gerota's fascia**, so the kidney does not fall onto the hilum; mobilise the colonic mesentery medial to the great vessels.

Proximal dissection for nephroureterectomy

- Identify the proximal ureter near the lower pole and dissect medially toward the renal vessels, avoiding overmanipulation and **keeping generous perihilar fat.**

- **Early ureteric ligation (clip or tie) is recommended**, as in open surgery.
- **Take a wide margin around the renal pelvis and hilum**, because lymphatic spread is a common route of progression — especially when advanced disease is suspected.
- Expose the vessels as medially as possible; ligate and divide the artery, then the vein, with a vascular-load stapler or multiple clips (Fig. 139.7); free the kidney outside Gerota's fascia; the adrenal is not removed routinely.
- **Ureteric blood supply**: anteromedial in the proximal third, medial in the middle third, lateral in the distal third.
- The lower ureter may require **port-hopping** or additional ports; in bulky ureteric disease include a wide margin. Continue as far as feasible and complete the remainder through the lower abdominal extraction incision.
- Specimen in a retrieval bag; inspect the bed for bleeding and lymphostasis. For open distal ureterectomy, rotate the table and make a Gibson, Pfannenstiel or low midline incision. **Close the fascia at ports of 10 mm and larger.**

Laparoscopic dissection of the distal ureter and bladder cuff

- **The open principles apply**: complete resection of the orifice and avoidance of urine and tumour spillage.
- **Sequence**: Trendelenburg position (bowel out of the pelvis) → extra ports as needed → extend the peritoneal incision from the iliac vessels into the perivesical space → free the ureter below the iliac vessels, following it medial to the umbilical ligament (Fig. 139.8; nodal tissue may be stripped off the vessels and left en bloc with the ureter) → divide the vas deferens (male) or round ligament with medial mobilisation of ovary and tube (female) → trace the ureter between bladder and medial umbilical ligament, under the superior vesical artery, to the detrusor → **clip or tie the ureter at the detrusor** (Fig. 139.9) → divide the lateral bladder pedicle to rotate the bladder medially and expose the whole intramural ureter → partial filling aids visualisation → free the ureter extravesically by dissecting the detrusor down to the urothelium around the tunnel → **empty the bladder and confirm catheter function** → open the bladder and complete excision under direct vision of the orifice.
- **Alternative in a complete extravesical approach**: flexible cystoscopy confirms complete ureterectomy and patency of the contralateral orifice.
- Multilayer absorbable closure as in open surgery; perivesical drain after watertight closure is confirmed by irrigation.

Robot-assisted laparoscopic nephroureterectomy

- **A feasible alternative to open and pure laparoscopic surgery**; meta-analytic data suggest comparable outcomes with shorter length of stay for minimally invasive procedures.
- **Newer systems** (longer instruments, greater range of motion) allow the distal ureterectomy **without re-docking or repositioning**.
- **Port placement is paramount; a four-port configuration is common**:

	Si system	Xi system (Fig. 139.10)
Array	Camera at umbilical level lateral to the rectus sheath; 8-mm port 1 (cephalad) and port 2 (caudad) on the same line, 7–8 cm from the camera; port 3 about 5 cm cranial to the iliac crest near the anterior axillary line; assistant port in the midline at or around the umbilicus	Straight diagonal line from a cephalad midline port toward the iliac crest, 8-cm increments ; line angled from costal margin toward the pelvis

	Si system	Xi system (Fig. 139.10)
Arms	Left arm port 1, right arm port 2, fourth arm port 3 (retraction); for the distal ureter, retraction moves to port 1 and the left arm to port 3	Camera in the second port from the top, moved to the third for the cuff if needed; instruments swapped between ports as required
Pelvic phase	—	10–12° Trendelenburg

- **Extravesical dissection:** a distended bladder helps trace the ureterovesical junction; empty the bladder once the ureter is out of the detrusor; stay sutures medial and lateral to the ureterovesical incision aid reconstruction; **two-layer bladder closure**.

Outcomes and Comparative Data

- **First laparoscopic nephroureterectomy: Clayman, 1991.**
- Published series vary by approach (transperitoneal versus retroperitoneal) and by distal management (open removal, transurethral resection, totally laparoscopic); **no approach has a clear-cut advantage** in morbidity, cosmesis or return to activity.
- **The only randomised trial** of laparoscopic (LNU) versus open (ONU) nephroureterectomy was early and underpowered: **no significant difference in 5-year cancer-specific or metastasis-free survival**; blood loss and time to discharge significantly lower with LNU.

OPEN, LAPAROSCOPIC AND ROBOT-ASSISTED NEPHROURETERECTOMY COMPARED

	Open (ONU)	Laparoscopic (LNU)	Robot-assisted (RANU)
Oncologic outcome	Reference	Equivalent to ONU (overall, cancer-specific, recurrence-free survival), even stratified by stage and nodes; early data suggested worse recurrence-free survival in pT3/pT4 that longer follow-up did not confirm	Comparable to LNU (cancer-specific, recurrence-free and overall survival)
Perioperative morbidity	Higher	Lower; less blood loss, shorter stay, fewer complications	Lower blood loss, complications, stay and 30-day mortality than LNU ; lower margin rates in pooled data
Operative time	—	Shorter than RANU	Longer
Lymph node dissection	Fewer nodes than robotic	—	Node dissection performed more often; approach does not compromise yield
Caveats	—	Possibly more bladder recurrence; less favoured for T3/T4	Ergonomics, assisted retraction, articulated wrist for dissection and suturing; evidence retrospective

- **AUA/SUO: evidence of oncologic equivalence between ONU and LNU is sufficient to support either**, with shorter hospital stay and fewer complications for minimally invasive surgery at experienced centres.
- **RANU by experienced surgeons may offer superior surgical and oncologic outcomes over LNU** on retrospective evidence; prospective trials and long-term follow-up are needed to confirm.
- **Local recurrence and incision or port-site seeding** are concerns with any approach, because urothelial cancer implants after spillage. Reported cases include port or incision recurrences after simple nephrectomy for presumed benign disease with unsuspected high-grade UTUC (oncologic principles not followed), and a local recurrence after perioperative violation of the ipsilateral urinary tract. One small

series found comparable recurrence overall but **peritoneal carcinomatosis only after LNU**, suggesting that recurrence patterns may differ in minimally invasive cases when spillage occurs.

- **Summary:** no significant difference between laparoscopic-based and open RNU when oncologic principles are followed; the collecting system and bladder cuff must be handled carefully and completely.

Concept — *Access matters less than principle: open, laparoscopic and robotic RNU converge in oncologic outcome as long as the specimen stays closed, and the rare disasters — port-site and peritoneal seeding — trace back to spillage, not to the approach.*

Lymphadenectomy

- **Regional nodal metastasis is characteristic of urothelial carcinoma, and UTUC is no exception.** Lymphadenectomy at extirpation provides accurate staging and demonstrable therapeutic benefit; the open questions are extent of dissection and risk-adapted selection.
- **Templates are based on the anatomical site of the primary**, but overlap is substantial because tumour location and retroperitoneal drainage vary; **wide regional templates are therefore advocated.**
- **Classic templates:** renal pelvis, proximal or mid-ureter → ipsilateral hilar nodes plus para-aortic (left) or paracaval (right) nodes; lower ureter → pelvic nodes.
- **Extended templates** derive from node-mapping data and reflect **right-to-left lymphatic drainage** (Fig. 139.11):

EXTENDED LYMPHADENECTOMY TEMPLATES

Primary site	Right-sided	Left-sided	Caudal limit
Renal pelvis	Hilar, paracaval, retrocaval, interaortocaval	Hilar, para-aortic	Renal vessels → inferior mesenteric artery
Upper two-thirds of ureter (above the common iliac crossing)	As for renal pelvis	As for renal pelvis	Aortic bifurcation
Lower third of ureter	Ipsilateral common, external and internal iliac, obturator and presacral		Pelvic

- **Node mapping (multicentre):** for right-sided pelvic and proximal ureteric tumours, hilar, paracaval and retrocaval dissection captures most metastases and **adding interaortocaval dissection raises capture to near-complete**; for left-sided tumours, hilar and para-aortic dissection captures most and interaortocaval nodes add only modestly. Mid- and distal-ureteric patterns are less clearly defined.
- **Node positivity is 12–25% overall, rising with stage and grade:** pTa/pTis 0–3%, pT1 0–6.3%, pT2 0–40%, pT3 19–47%, pT4 20–100%. Database analyses are confounded by non-uniform templates and widely varying node counts.
- **Outcomes: pN0 better than pNx; pN+ worse than pNx.**
- **Node count is used as a surrogate for extent:** removal of **eight or more nodes** increased the chance of finding positive nodes and improved disease-specific survival in disease above pT1; higher yield independently predicts overall survival.
- **Approach does not compromise yield** (more nodes with robotic than open surgery in national data); **surgeon factors are the larger determinant of yield**, exceeding approach or pathological assessment.
- **Template lymphadenectomy independently predicted overall survival in muscle-invasive renal pelvis tumours;** other studies suggest a therapeutic benefit.
- **Complications:** overall rates modest and in line with open and laparoscopic series; the specific risk is **lymphatic or chylous leak.**

- **Summary:** lymphadenectomy is safe, stages accurately, and has prognostic and therapeutic value in **invasive disease (T2–T4), especially of the renal pelvis and proximal ureter**; selection and extent will be refined by genomics and imaging.

Concept — *Lymph drains right to left and top to bottom, so the template is set by side and level: right-sided upper tumours need the interaortocaval nodes, and ureteric tumours move the field caudally in step with the primary.*

Results of Nephroureterectomy

- **RNU is the gold standard for oncologic control of non-metastatic disease**, based on large retrospective series; complete resection gives durable local control and cure when disease is truly localised.
- **Bladder recurrence is common and treatable**, most likely from drop-down implantation (supported by genomic evidence). Outcome reporting must therefore **separate bladder recurrence from extra-urothelial recurrence** as an end point.
- **Surgical outcomes have been stable over three decades**: in a large multi-institutional review with minimal neoadjuvant use, disease-free and cancer-specific survival did not change by stage and grade. **Organ-confined disease ($\leq$ pT2, N0/Nx) had consistently good 5-year cancer-specific survival; non-organ-confined disease ($\geq$ pT3 and/or N+) less than half of that** — the baseline for surgery alone, underlining stage-dependent lethality and the need for adjunctive therapy.
- **Loss of renal function is the key cost**; in poor renal function, the curative value of RNU is balanced against the morbidity of dialysis.
- **Kidney-sparing versus RNU**: a meta-analysis of retrospective comparative studies found comparable overall and cancer-specific survival after adjustment, but **about three times as many reinterventions with endoscopic management**, reflecting upper-tract recurrence. Selection bias is strong; the cumulative burden must be explained to patients. **High kidney-preservation rates are achievable in well-selected cases** with careful technique.

Concept — *RNU buys certainty of local control at the price of nephrons; every kidney-sparing strategy trades that certainty for renal function and repays it in reinterventions and surveillance.*

Segmental Ureteral Resection

- **Segmental ureterectomy here means mainly distal ureterectomy with reimplantation.** True proximal segmental resection with uretero-ureterostomy is rarely indicated; total ureterectomy may require ileal ureter or autotransplantation.
- Comparative data are retrospective with marked selection bias; data quality is poor.
- **Resection for compromised renal function must not sacrifice oncologic efficacy to avoid dialysis.**
- **Ureteroscopy of the entire upper tract above the tumour is critical to exclude multifocality.**
- **High-grade disease → many surgeons add ipsilateral pelvic lymph node dissection.**
- **Advantages:** preserves renal function and yields a pathological specimen that informs the need for, and extent of, adjuvant therapy.

Open Segmental Ureterectomy

Ureteroureterostomy

- **Indications:** tumours of the proximal or mid-ureter not removable endoscopically, or high-grade or invasive tumours when the renal unit must be preserved. **A clear margin with enough well-vascularised ureter for a tension-free anastomosis is paramount and is the main limiting challenge.** The procedure is rare and narrowly indicated: a small tumour, no multifocality elsewhere, not endoscopically manageable.

- **Technique:** full or modified flank position → flank incision from the tip of the 12th rib → extraperitoneal approach; ureter identified, mobilised and slung in vessel loops → tumour palpated; **ureter ligated 1–2 cm above and below the tumour** (location verifiable on preoperative imaging; Fig. 139.12) → excision; **margins confirmed by frozen section** → regional lymphadenectomy → both ends spatulated and anastomosed with **interrupted 4-0 polyglactin (Vicryl)** → ureteric stent placed before completion → drain at the surgeon's discretion.
- **Success depends on preserving ureteric blood supply and a tension-free anastomosis;** after a long excision, mobilise and descend the kidney for length.
- **Laparoscopic or robotic:** flank position, trocars as for pyeloplasty or nephrectomy; strict oncologic principles to avoid tumour violation and obtain negative margins.

Distal ureterectomy and direct neocystostomy, or ureteroneocystostomy with psoas hitch or Boari flap

- Distal ureterectomy as described above; the **entire distal ureter and bladder cuff must be excised** and the posterior cystotomy closed in two layers. Laparoscopic or robotic: dorsal lithotomy or supine with Trendelenburg, as for prostatectomy. The ureter is mobilised for a tension-free anastomosis and spatulated.

Non-refluxing anastomosis	Refluxing anastomosis
Confines infection to the lower tract; theoretically avoids seeding the upper tract	Easier upper-tract surveillance
• The choice remains a matter of debate. • Extravesical reimplant: incise detrusor to expose mucosa → mucosal slit at the distal aspect → full-thickness ureter-to-mucosa anastomosis, continuous or interrupted 3-0 polyglactin or polydioxanone → two distal sutures through the full thickness of the bladder wall anchor the ureter against sliding out → detrusor closed over the ureter with interrupted absorbable sutures (for example 2-0 polyglactin) to create a non-refluxing mechanism → stent optional. • Intravesical reimplant: anterior cystotomy → incision in the posterolateral wall → 2–3-cm submucosal tunnel → ureter brought through, spatulated and anastomosed with interrupted absorbable sutures. • Psoas hitch — about 5 cm extra length: mobilise the bladder anteriorly and laterally; divide the round ligament in females; the contralateral superior vesical artery and lateral pedicle may be divided for mobility; suture the ipsilateral dome to the psoas tendon with several interrupted sutures, avoiding nerve injury or entrapment. • Boari flap — a further 10–15 cm, sometimes reaching the renal pelvis (Fig. 139.13). Preoperative cystogram to assess capacity: a small-capacity bladder, especially if irradiated, is a contraindication. U-shaped flap (L-shaped for more length); base at least 2 cm wider than the apex (blood supply); width at least three times the ureteric diameter (tube calibre); tip fixed to the psoas; spatulated ureter anastomosed end to end; flap tubularised with two-layer absorbable closure over a ureteric catheter. • After any of these: retroperitoneal suction drain and 5–7 days of Foley drainage; cystogram before catheter removal after extensive reconstruction. • Fig. 139.13 shows subtotal ureterectomy for multiple diffuse tumours, a spiral flap from the anterior bladder wall, and psoas hitch plus Boari flap reaching the remaining proximal ureter.	

Ileal Ureteral Replacement

- **A long diseased segment is replaced with ileum;** the appendix has also been used for segmental substitution.
- **Technique:** midline intraperitoneal incision → harvest **20–25 cm of ileum at least 15 cm from the ileocaecal valve** → stapled bowel anastomosis → proximal end-to-end anastomosis to the renal pelvis, **isoperistaltic**, running absorbable suture (end-to-side to a healthy proximal ureter if present) → stent →

distal end-to-side anastomosis to the posterior bladder wall through an intravesical approach, two layers → retroperitoneal suction drain near the anastomoses → **large Foley for at least 1 week**, irrigated frequently if needed; nephrostomy optional → **cystogram and nephrostogram before tube removal**.

- **Alternatives: renal autotransplantation** in skilled hands; or **renal mobilisation with nephropexy** of Gerota's fascia to the cut edge of the peritoneum (or psoas tendon), with caudal traction — **up to 8–10 cm on the left**, because the left renal vein is longer (Fig. 139.14); performed laparoscopically, avoiding a second flank incision.

GAINING URETERIC LENGTH

Manoeuvre	Length gained	Key condition
Psoas hitch	About 5 cm	Mobile bladder
Boari flap (with hitch)	Further 10–15 cm; may reach pelvis	Adequate capacity; small or irradiated bladder contraindicates
Renal descensus with nephropexy	Up to 8–10 cm	Left side (longer renal vein)
Ileal ureter	Whole ureter	20–25 cm ileum, ≥15 cm from ileocaecal valve
Autotransplantation	Whole ureter	Skilled hands; alternative to ileal ureter

Laparoscopic or Robotic Distal Ureterectomy and Reimplantation

- **Indications as for open surgery; reserved for low-risk distal tumours.** The robot assists the reconstruction. The ureter is dissected down to the orifice and the proximal end anastomosed to the bladder by standard techniques. Early reports are encouraging; strict oncologic principles apply.

Results

- **Historical poles:** RNU for all UTUC, versus segmental ureterectomy only for low-grade, non-invasive distal ureteric tumours.
- **Ureteric cancers occur most often in the distal ureter, then mid-ureter, then proximal ureter.**
- **In properly selected patients, segmental resection gives oncologic outcomes similar to RNU;** outcome correlates with stage and grade regardless of the extent of surgery (Table 139.1).
- **Series bottom lines:** after adjustment, segmental ureterectomy and RNU gave similar outcomes (mostly non-muscle-invasive disease, long follow-up); recurrences after organ-sparing surgery occurred in invasive disease; low-grade non-invasive tumours treated segmentally had minimal recurrence and no cancer deaths; T3 disease had very poor cancer-specific survival; a Surveillance, Epidemiology, and End Results (SEER) population analysis showed no difference in cancer-specific mortality, segmental versus RNU, adjusted for stage; a single-institution comparison found no difference in bladder, local or distant recurrence or survival.
- **Ipsilateral recurrence after conservative treatment of ureteric tumours is 33–55%,** mostly distal to the original lesion but also proximal. **Risk is lifelong because late recurrences occur,** so follow-up is lifelong. These figures come from an era of limited ureteroscopy; thorough ureteroscopic evaluation and selection probably improve results.
- **Segmental ureterectomy** → low-grade, non-muscle-invasive disease of the **proximal or mid-ureter** not completely ablatable endoscopically (size or multiplicity).
- **Distal ureterectomy with neocystostomy** → low-grade, low-stage, or selected high-grade locally invasive **distal** tumours when renal preservation is necessary. With high-grade disease, consider lymphadenectomy.

TABLE 139.1 LITERATURE REVIEW OF OVERALL SURVIVAL OF PATIENTS WITH UPPER TRACT UROTHELIAL TUMOURS (RENAL PELVIS OR URETER) BY STAGE AND GRADE

Category	5-year survival (%)
Grade 1–2	40–87
Grade 3–4	0–33
Ta, T1, Tis	60–90
T2	43–75
T3	16–33
T4	0–5
N+	0–4
M+	0

TNM, tumour, node, metastasis.

- **The prognostic cliff lies between T2 and T3** — invasion beyond the muscularis roughly halves survival.
- **High grade (3–4) carries survival similar to T3**, which is why grade stands in for stage in risk stratification.
- **Node-positive disease behaves almost like metastatic disease** with surgery alone — the rationale for systemic therapy.

Concept — *In the ureter the pathology, not the operation, sets the prognosis: once selection has excluded multifocality, segmental resection matches RNU, so the work lies in selection (full ureteroscopic mapping) and in lifelong surveillance of what is left.*

ENDOUROLOGIC MANAGEMENT

Basic Attributes

- **Hugh Hampton Young: first endoscopic evaluation of the upper urinary tract, 1912.**
- **Miniaturisation** allows biopsy and ablation through the smallest instruments and makes surveillance practical with small ureteroscopes, usually **without pre-stenting or active dilation** of the distal ureter.

Retrograde ureteroscopic approach	Antegrade percutaneous approach
Low-volume ureteric and renal tumours	Larger tumours of the upper ureter or kidney ; tumours hard to reach retrogradely (e.g., lower-pole calyx); previous urinary diversion

- **Multifocal disease → combined antegrade and retrograde approaches** (Fig. 139.15: synchronous bilateral tumours — right renal cell carcinoma needing radical nephrectomy, left proximal ureteric tumour treated by combined ablation).
- **Principles mirror the bladder**: sample the tumour, then ablate with electrocautery or laser.
- **Grade is the key selection tool**: endoscopic treatment is safe and feasible for verified low-grade disease; results for high-grade tumours are poor.
- **Staged procedure for high-volume disease or suspected high grade or stage**: when RNU will probably be needed, perform **only biopsy and partial ablation** to minimise perforation and major complications; complete endoscopic management only after pathology confirms suitability.
- **Unresectable, high-grade or invasive disease → proceed immediately to RNU**; segmental resection is an option in a solitary kidney or significant renal insufficiency if the patient is medically fit.
- **Renal-sparing therapy requires a lifetime commitment to radiographic and endoscopic follow-up.**

Retrograde Ureteroscopic

Ureteroscopy and Ureteropyeloscopy

- **First described for tumours by Goodman, 1984**; favoured for ureteric and smaller renal tumours. Small-diameter rigid and flexible scopes have made tumour location less limiting.

Advantages	Disadvantages (all from small instruments)
Lower morbidity than percutaneous or open surgery; closed system — non-urothelial surfaces are not exposed to seeding	Small field of view and working channel limit the tumour size treatable; lower-pole calyces unreliable ; cannot remove large tumours or obtain deep specimens for staging ; difficult after urinary diversion

Technique and instrumentation

	Rigid ureteroscope (Fig. 139.17A)	Flexible ureteropyeloscope (Fig. 139.17B)
Territory	Distal and mid-ureter ; upper ureter and kidney unreliable, especially in males	Upper ureter and kidney ; passes to most of the tract
Size	Larger scopes: better view and irrigation; small (8 Fr) scopes usually need no active dilation	Available below 8 Fr
Limitations	Reach	Small working channel limits irrigant flow and instrument diameter; lower pole limited by the infundibulopelvic angle ; prior diversion

Endoscopic evaluation and collection of urine cytology

- **Sequence**: cystoscopy (bladder inspected; concomitant disease addressed as indicated) → instrumented cytology if not already obtained → inspect the orifice for **lateralising haematuria** → atraumatic guidewire into the lower ureter, or a **no-touch** direct-access approach → small-diameter (6.9- or 7.5-Fr) ureteroscope; inspect the distal ureter for wire or dilation trauma → advance under vision from the ureterovesical junction to the renal pelvis → flexible scope for complete pelvicalyceal inspection → **map lesions** (often with fluoroscopy) and document appearance and size → **saline washing of a suspicious area improves the diagnostic yield of cytology**.
- **Tight ureter**: gentle minimal dilation, or **pre-stenting for delayed interval ureteroscopy** (passive dilation) to minimise injury or perforation.
- **Ureteric access sheaths** — **no recommendation either way**:

For access sheaths	Against access sheaths
Low-pressure system; repeated access; larger specimens removed with less theoretical downstream seeding (lower ureter bypassed)	Additional orifice dilation; reduced scope flexibility near the ureteropelvic junction; need for postprocedure stenting; risk of significant ureteric injury

Tumour Confined to the Intramural Ureter

- **Tumour protruding from the orifice** → **complete ureteroscopic ablation, or aggressive transurethral resection of the entire most distal ureter**, with acceptable results.

Biopsy and definitive treatment; ureteroscopic techniques

- **Specimens are minute**: place in fixative at once and label specifically for histological or cytological evaluation.

URETEROSCOPIC TECHNIQUES (FIG. 139.18)

Technique	Method	Best use / caution
Bulk excision with ablation of base	Debulk with biopsy forceps, or flat wire basket engaged alongside the tumour (avoid avulsing the ureter); base treated with electrocautery or laser	Low-grade papillary tumour on a narrow stalk
Resection to base	Ureteroscopic resectoscope; intraluminal tumour only — no attempt to resect beyond the lamina propria , unlike bladder resection	Mid- and upper ureter thin, prone to perforation; large distal disease: extensive dilation then long standard resectoscope
Biopsy, then ablation	Forceps sampling (multiple bites with small 3-Fr flexible forceps), then bulk ablated to its base with laser or electrosurgical energy	General approach; laser preferred — more reliable energy delivery and depth of penetration

- **Bugbee electrode (2 or 3 Fr) fulguration:** variable depth of penetration makes it dangerous in the ureter; **avoid circumferential fulguration — high stricture risk.**
- **Laser fibres (200 or 365 µm) pass through flexible scopes** without significant loss of irrigant flow or deflection.

LASER ENERGY SOURCES FOR UPPER TRACT TUMOUR ABLATION (FIG. 139.19)

	Holmium:YAG (Ho:YAG)	Neodymium:YAG (Nd:YAG)	Thulium (fibre laser; Tm:YAG)
Tissue penetration	<0.5 mm	Up to 5–6 mm	<0.2 mm
Mechanism	Ablation with good haemostasis	Coagulative necrosis , then sloughing of necrotic tumour	Photovaporisation / photothermolysis; strong coagulation
Typical settings	0.6–1 J at 10 Hz (varies with fibre and pulse duration)	15 W × 2 s ablation; 5–10 W × 2 s coagulation	1 J at 10 Hz; longer pulse for coagulation; quasi-continuous or SuperPulsed modes for haemostasis
Wavelength	—	—	1940 nm (fibre) / 2010 nm (Tm:YAG), near the water-absorption peak; absorption several times that of Ho:YAG
Role	Well suited to the ureter; cumbersome for large renal pelvis tumours	Low safety margin in the thin ureter (transmural thermal injury)	Early experience favourable; recurrences frequent; complications mainly low grade; comparative data awaited

YAG, yttrium-aluminium-garnet.

- **Recurrence is expected after ablation**, as in the bladder; **several procedures over months** may be needed for complete response, especially with larger tumours.
- **Stenting after treatment is at surgeon discretion;** pre-stenting in difficult access reduces ureteric injury and tumour spillage.
- After eradication: surveillance with imaging, urinary cytology and endoscopy. **AUA/SUO: do not reflexively perform invasive endoscopy of a clinically normal contralateral upper tract.** Long-term follow-up for all, adjusted to sites and intervals of recurrence.

Results

- **No randomised trials compare endoscopic therapy with RNU — all evidence is case series (level 4).**
- **Large literature review:** upper-tract recurrence in about half, bladder recurrence in about a third; progression and metastatic recurrence in a minority; ureteroscopic therapy failed in about a quarter, most proceeding to RNU — with strong bias toward favourable tumours (unifocal, low grade, small).

- **Grade is the most important predictor of recurrence;** upper-tract recurrence and disease-free survival worsen with higher grade.
- **Ipsilateral recurrence is the dominant concern:** one series found recurrence in nearly all patients, with several recurrences each, and 5-year recurrence-free survival is modest — important when the contralateral kidney is normal. Counsel for lifelong follow-up and retreatment.
- **Complications are uncommon,** usually related to comorbidity; ureteroscopy-specific complications include **perforation (managed with a stent) and stricture**, and have fallen with smaller scopes, better lasers and refined technique.

TABLE 139.2 URETEROSCOPIC MANAGEMENT — RANGE ACROSS SEVEN SERIES

End point	Range
Follow-up	18–55 months
Upper tract recurrence	23–90%
Bladder recurrence	15–45%
Nephroureterectomy	3–33%
Disease progression	0–20%
Failed management	3–47%

Modified from Cutress ML, Stewart GD, Zakikhani P, et al.: Ureteroscopic and percutaneous management of UTUC: systematic review, BJU Int 110:614–628, 2012.

- **Recurrence dominates, progression is uncommon:** ureteroscopic management trades frequent reintervention for kidney preservation. The wide ranges reflect selection.
- **Case selection, with technical experience, is the overriding factor:** size, multifocality, vascularity, location and grade. **Tumours of the renal papillae or fornix** are hard to treat without perforation, bleeding or functional damage. **High-grade and invasive tumours are not ideal** except in imperative settings (solitary kidney, small individual tumours).
- **Accuracy of ureteroscopic sampling:** diagnostic accuracy is high, but grade concordance with the RNU specimen is variable; **adding instrumented selective upper-tract cytology to biopsy improves grading accuracy.** Genomic markers from biopsy and selective cytology show high concordance with the primary and predict grade and stage.
- **Grade predicts stage:** most grade 1–2 tumours are non-invasive (Ta/T1); most grade 3 tumours are invasive (T2/T3). Stage cannot be assessed directly, but non-invasive disease can be expected in most low-grade tumours.
- **Does ureteroscopy spread tumour?** Tumour has appeared in refluxing ureters and after ureteroscopic treatment; a meta-analysis found **higher bladder recurrence after prior ureteroscopic tumour manipulation**, strongly supporting downstream seeding. Mitigation: topical intracavitary therapy after treatment and access sheaths (each with its own risks). Tumour cells in renal lymphatics after ureteroscopy raised concern about metastasis, but **ureteroscopy before RNU has not increased metastatic risk.**

Concept — Ureteroscopic management is a closed-system, low-morbidity strategy whose weakness is information: it cannot stage, so everything depends on grade — grade selects the patient, predicts the stage, and predicts the recurrence that will follow.

Antegrade Nephroureteroscopic

- **First described by Tomera, 1982;** favoured for **larger tumours of the renal pelvis or proximal ureter**, tumours difficult to reach retrogradely, and **urinary diversions.**

- **Main advantage: larger instruments** remove a large tumour volume anywhere in the collecting system, and the upper tract can be reached when retrograde access is technically difficult.
- **AUA/SUO support endoscopic therapy for tumours <1.5 cm** (lower risk of invasion); **caution for tumours ≥1.5 cm** (elevated risk of invasive disease).
- **Deeper biopsies facilitate staging and grading**; the approach avoids flexible-ureteroscope limits (complex calyces, lower-pole calyx, diversion); the tract is kept for **immediate postoperative nephroscopy and topical adjuvant therapy**.
- **Disadvantages**: greater morbidity than ureteroscopy; inherent tract risks; usually inpatient admission; an **open system** — loss of urothelial integrity exposes non-urothelial surfaces, allowing **tract implantation**.

RETROGRADE VERSUS ANTEGRADE ENDOSCOPIC MANAGEMENT

	Retrograde ureteroscopic	Antegrade percutaneous
First described	Goodman, 1984	Tomera, 1982
System	Closed	Open
Tumour volume	Low	Large
Staging	Shallow biopsies — cannot stage	Deep biopsies, base sampled — can stage
Morbidity	Lower	Higher; admission
Seeding route	Downstream (bladder)	Nephrostomy tract (rare)
Pooled recurrence	Higher bladder and upper-tract recurrence	Lower bladder and upper-tract recurrence
Second look / topical therapy	Repeat ureteroscopy; ureteric catheter or stent reflux	Through the retained nephrostomy

Technique and Instrumentation

Establishment of the nephrostomy tract

- **Sequence**: cystoscopy → open-ended ureteric catheter into the pelvis → contrast to define calyceal anatomy → percutaneous puncture of the chosen calyx (Fig. 139.20). In the prone split-leg position, a flexible ureteroscope can be passed to the target and access obtained under direct and fluoroscopic guidance.

CHOICE OF PUNCTURE (FIG. 139.20)

Tumour site	Puncture
Peripheral calyx	Direct puncture as far distally in the calyx as possible , distal to the tumour — avoid traumatising or puncturing the tumour (Fig. 139.21)
Renal pelvis, upper ureter	Upper or middle calyx — allows manoeuvring through the pelvis and down the ureteropelvic junction
Lower calyx	Lower-calyx puncture

- **Tract dilation**: sequential (Amplatz) or balloon, to a **30-Fr sheath**. Correct tract position is crucial — by the urologist, or by the radiologist after direct consultation with the operating surgeon.
- **Two-stage option**: tract first, allowed to mature for **1–2 weeks**, then dilation and treatment.
- **Diagnostic-only procedures** (e.g., positive cytology after cystectomy and diversion): the initial small tract admits a flexible ureteroscope.

- **Otherwise:** nephroscope inserted → ureteric catheter grasped, brought out through the tract and exchanged for a stiff guidewire (antegrade and retrograde control) → complete rigid and flexible nephroscopy → **any suspicion of upper ureteric involvement warrants antegrade ureteroscopy.**

Biopsy and definitive therapy

PERCUTANEOUS TECHNIQUES (FIG. 139.22)

Technique	Method	Best use / caution
Cold-cup forceps	Standard nephroscope; bulk grasped and removed piecemeal to the base; separate base biopsy for staging ; base cauterised with a Bugbee electrode	Low-grade papillary lesions on a thin stalk (minimal bleeding); vascular or bulky tumours bleed and obscure the view
Resection loop	Standard or bipolar resectoscope to the base; base resected and sent separately for staging	Larger, broad-based tumours ; monopolar risks absorption of large volumes of hypo-osmotic irrigant, so bipolar preferred ; avoid resection into major renal vessels
Laser	Rigid or flexible scope; sample, then Ho:YAG or Nd:YAG (thulium also effective); tissue may be taken with a small gastrointestinal polyp snare	Through a standard nephroscope or a flexible cystoscope

- **A nephrostomy tube is left in all cases**, allowing **second-look nephroscopy** to confirm complete removal (Fig. 139.23: solitary kidney with a 5-cm renal pelvis tumour treated by staged resection; grade 1 without submucosal invasion).
- **RNU is indicated if pathology shows high-grade or invasive disease.**

Results

- **No randomised trials; only limited retrospective series** (Table 139.3).
- **Meta-analysis of contemporary antegrade series: pooled bladder and upper-tract recurrence lower than after retrograde surgery.** Tumour grade strongly predicts outcome.
- **AUA/SUO risk-adapted kidney preservation:** ablation **strongly recommended in low-risk favourable** disease and an **option in low-risk unfavourable** disease; in **high-risk favourable** disease, forgoing RNU is highly selected and individualised (low-volume tumours, or patients unable to undergo RNU), with higher progression risk and a **low threshold for definitive resection.**
- **Complications** resemble those of benign percutaneous surgery: bleeding, systemic absorption of hypo-osmotic irrigant (monopolar resection), collecting-system perforation, secondary ureteropelvic junction obstruction. Overall complication rate about a quarter; the most significant are transfusion, dialysis and renal failure. **Complications increase in number and severity with tumour grade** (more extensive disease and treatment).
- **Unlike ureteroscopic resection, the percutaneous method can stage tumours;** stage rises with grade.
- **Tract seeding:** few reported cases, all high grade (urothelial and squamous histology); overall rate about 0.3% — **possible but uncommon.**

TABLE 139.3 PERCUTANEOUS MANAGEMENT — RANGE ACROSS SIX SERIES

End point	Range
Follow-up	45–64 months
Upper tract recurrence	13–65%
Bladder recurrence	15–42%
Nephroureterectomy	13–50%

End point	Range
Disease progression	8–35%
Failed management	23–50%

Modified from Cutress ML, Stewart GD, Zakikhani P, et al.: Ureteroscopic and percutaneous management of UTUC: systematic review, BJU Int 110:614–628, 2012.

- **Against Table 139.2, percutaneous series show more nephroureterectomy and progression** — they treat larger, higher-grade tumours — while recurrence remains frequent in both.

Consideration for Urinary Diversions

- **The ureteroenteric anastomosis is hard to find**, and antegrade passage of a guidewire may be needed before endoscopy.
- **Conduits and neobladders**: a trial of flexible endoscopy into the diversion is reasonable. **Know the anastomosis type** — conduit: Wallace versus Bricker (determines where the ureters lie); neobladder: pouch anatomy (e.g., Studer).
- **Ureters not identifiable** → antegrade wire into the pouch (urologist or interventional radiology) → wire retrieved from the diversion → retrograde ureteroscopy over it; the tract need not be fully dilated. If this fails → tract dilation and antegrade uretero-nephroscopy.

Concept — *The percutaneous route exchanges the closed system for size and depth: it removes more, stages better and recurs less, at the price of morbidity and a tract that could seed — so it is reserved for tumours too big, too proximal or too inaccessible for the ureteroscope.*

Management of Positive Upper Tract Urinary Cytology and Carcinoma in Situ

Evaluation

- **An unequivocal positive voided cytology usually indicates urothelial carcinoma, most often from the bladder**; extravescial sources include the upper tracts and, in males, the prostatic urethra.
- Diagnosis is difficult: imaging of the upper tracts is limited, upper-tract endoscopy is more complex than cystoscopy, and minute specimens hamper histological diagnosis and staging.
- **Algorithm (Fig. 139.24): repeat cytology to confirm** → upper-tract imaging (usually CT urography) and **complete bladder evaluation** — bladder biopsies with or without enhanced visualisation, and resection of any tumour → **bladder positive**: treat the bladder (intravesical therapy and/or resection) and follow voided cytology → **cytology still positive** despite a negative bladder evaluation, or after adequate bladder treatment (after a "6 + 1" bacillus Calmette-Guérin [BCG] course) → **extravesical evaluation**, with repeat bladder evaluation at the same time.
- **Extravesical evaluation**: selective cytology from each upper tract, avoiding contamination from bladder or urethra; retrograde pyelograms if the upper tracts were not evaluated or visualised on CT urography; resection (biopsy) of a representative prostatic urethral specimen in males; ureteroscopy now, or once the positive cytology is lateralised.

Carcinoma in Situ of the Upper Urinary Tracts

- **Isolated upper-tract carcinoma in situ (CIS) is hard to diagnose** because adequate urothelial tissue cannot be obtained. Enhanced imaging (narrow-band or blue-light) and endoscopic brush biopsy help when cytology or standard biopsy is equivocal (Fig. 139.25).
- **It is often a diagnosis of exclusion: persistently positive selective cytology with no ureteroscopic or radiographic finding.**

Isolated upper-tract CIS	CIS coexisting with high-grade papillary UTUC
Treatment not well established; topical, field-directed therapy ; surgery only with evidence of more than surface disease	Radical and possibly multimodal treatment — associated with significantly worse recurrence-free and cancer-specific survival

- **RNU for a unilateral cytological abnormality was past practice and is not recommended now.** Upper-tract cytology shares the specificity limits of bladder cytology; samples are small with low cell counts; **inflammation (infection, calculus) causes false positives**; a **later contralateral abnormality is not rare** after true-positive early CIS.
- **AUA/SUO: pelvicalyceal therapy with a 6-week induction course of topical BCG is an option for upper-tract CIS.** Series are limited, retrograde or antegrade, with variable response; definitions of detection and success are inconsistent. CIS appears to respond about as well as its bladder counterpart, with complete response reported across a wide range (41–100%).
- **A nephrostomy tube is the more reliable delivery route**; alternatively, weekly cystoscopy with a ureteric catheter placed in the upper calyx. Fig. 139.25: CIS in a solitary kidney treated by superficial laser treatment of visible sites followed by BCG via nephrostomy.
- **Most would not operate initially without a histological, radiographic or endoscopic finding**, because cytology alone gives false positives and future bilateral disease is likely. **Segmental resection is usually ineffective — field effect. RNU is indicated when radiography or endoscopy confirms more than surface disease.**
- **Frequent re-evaluation:** urinalysis, bladder and possibly selective cytology, cystoscopy, and retrograde pyelography or ureteropyeloscopy.
- **CIS at the ureteric margin during radical cystectomy:** management is controversial. It confers a risk of progression but recurrence may take long to appear; many never progress, and recurrences may arise at the margin or elsewhere, with a high risk of progression to death from metastasis in some. Intraoperative frozen sections are positive in a minority; **conversion to a negative margin is achieved in most and is associated with improved survival.**

Concept — Upper-tract CIS is defined by what cannot be seen: the diagnosis rests on cytology, which is fallible and field-wide, so treatment is topical and field-directed, and extirpation is reserved for proof of more than surface disease.

Expectant Management

- **Little evidence;** UTUC was not considered suitable for surveillance because of progression risk (especially high-risk disease), flank pain, chronic haematuria and obstruction.
- **Untreated UTUC has markedly worse overall survival than treated disease**, so surveillance suits only patients with very short life expectancy.
- **AUA/SUO: watchful waiting or surveillance is an option for selected patients** with significant comorbidity, competing mortality risks, or significant risk of end-stage renal disease with intervention — after a well-informed shared discussion with patient and family.

Adjuvant Topical Therapy

Adjuvant intravesical chemotherapy after nephroureterectomy

- **Bladder cancer develops in 30–50% after RNU and after endoscopic biopsy or treatment**; most bladder recurrences result from seeding and **occur within 2 years.**

- Mitomycin C (MMC) and, most recently, gemcitabine reduce recurrence after transurethral resection of bladder tumour (TURBT). **Two randomised trials show that a single postoperative dose of topical chemotherapy reduces bladder recurrence after RNU:**

RANDOMISED TRIALS OF SINGLE-DOSE INTRAVESICAL CHEMOTHERAPY AFTER NEPHROURETERECTOMY

Trial	Agent and timing	Bladder recurrence, control → treated	Design note
ODMIT-C (O'Brien, 2011)	MMC 40 mg in 40 mL saline, single dose at catheter removal	28% → 16% at 1 year	Biopsy confirmation of recurrence not required
Pirarubicin trial (Ito, 2013)	Pirarubicin, single dose within 5 days	42% → 17% at 2 years	Biopsy confirmation required

- **Despite different designs and drugs, results were remarkably similar at 1 year; both had very low adverse-effect rates** and good tolerance.
- **No UTUC evidence favours one agent**, but **gemcitabine** has become preferred in non-muscle-invasive bladder cancer (NMIBC), with meta-analytic superiority over MMC.
- **AUA/SUO strongly recommend a single perioperative dose of topical chemotherapy in eligible patients after RNU and after endoscopic tumour management.**

Technique

Postoperative instillation (3–7 days)	Intraoperative instillation (authors' preference)
Catheter still in place → cystogram confirms no leak at the cuff site → single dose of MMC, gemcitabine or other proven agent → catheter clamped, dwell 1–2 hours	Recurrence reduction comparable to postoperative instillation. Rationale: (1) no return for cystogram or extra visit; (2) earlier catheter removal; (3) no cancellation because of cuff leak; (4) given, theoretically, when seeding occurs

- **Intraoperative sequence:** two- or three-way catheter (on or off the sterile field) → drain bladder → instil chemotherapy before surgery starts → clamp outflow (and inflow if three-way) → anaesthetist minimises intravenous fluid (bladder clamped 1–2 hours) → **once the ureter is clipped below the tumour and after 1–2 hours' dwell**, unclamp and drain → the bladder is empty by the time dissection reaches the distal ureter and cuff; a three-way catheter allows irrigation to wash out residual drug and, after cystotomy closure, confirmation of a watertight repair.

After organ-sparing therapy

- **Any procedure short of extirpation carries higher local recurrence** (established ipsilateral risk); topical chemotherapy in the bladder or upper tract is used to reduce it.
- **Intravesical therapy after endoscopic management of UTUC is optional (AUA/SUO)**, extrapolated from immediate instillation after TURBT for NMIBC.
- **Topical therapy in the upper urinary tract:** for primary ablation or as adjuvant therapy after endoscopic or organ-sparing treatment. **Contact time is limited by gravity-dependent drainage.**

UPPER-TRACT INSTILLATION ROUTES CONSIDERED RELIABLE BY THE AUA/SUO

Route	Method
Antegrade	Through a nephrostomy tube (Fig. 139.26: gravity instillation with a pressure-limiting column of 25 cm H ₂ O)
Retrograde	Directly via a ureteric catheter

Route	Method
Intravesical with reflux	Indwelling ureteric stent to induce iatrogenic reflux

- **Whatever the route: low pressure and no active infection**, to minimise bacterial sepsis and systemic absorption; high pressure is linked to both.
- **Results:** the agents used in the bladder are used in the upper tract; historical studies are small, retrospective, uncontrolled series — thiotepa, MMC, gemcitabine, docetaxel, epirubicin, doxorubicin (Adriamycin) and BCG (Table 139.4). **Indications for pelvicalyceal therapy: solitary kidney, bilateral UTUC, or risk of progression to end-stage renal disease.**

TABLE 139.4 ADJUVANT UPPER TRACT INSTILLATION — BY AGENT

Agent	Series	Benefit	Serious complications
Thiotepa	2 (very small)	Not evaluated or none	Death from sepsis
MMC	4	Safety only or none in most; one series lower recurrence than without MMC	Deaths from systemic absorption (one series)
BCG	4	No benefit shown in any	Deaths from sepsis (two series)

Modified from Cutress ML, Stewart GD, Zakikhani P, et al.: Ureteroscopic and percutaneous management of UTUC: systematic review, BJU Int 110:614–628, 2012.

- **The table tests one message: adjuvant upper-tract instillation after papillary ablation has no proven benefit and carries rare fatal toxicity** (sepsis, systemic absorption) — hence low pressure and exclusion of infection.
- **BCG for primary CIS via nephrostomy gives favourable responses** (largest experience, with durable recurrence-free survival in a majority at 5 years); **adjuvant BCG after ablation of papillary tumours did worse**. With intrarenal BCG perfusion, cytology often normalises initially, but a substantial proportion recur and those recurrences were lethal — **initial response is not durable cure**. Removing a renal unit for CIS alone is not urged, but follow-up must be vigilant.
- **Commonest complications of instillation: bacterial sepsis and stricture**. Exclude active infection before every treatment and use only low-pressure delivery. Stricture may relate to prior endoscopy, ureteric catheterisation, the agent, or all three.
- **BCG-specific:** renal granulomatous involvement without systemic BCG infection (most common); disproportionate loss of renal function in patients with vesicoureteric reflux.
- **Adjuvant upper-tract BCG is an acceptable option after complete ablation of high-risk favourable Ta/T1 UTUC**; meta-analysis showed recurrence in a substantial minority with high cancer-specific survival.
- **UGN-101 (Jelmyto) — mitomycin-containing reverse thermal hydrogel, US Food and Drug Administration (FDA) approved 2020 for chemoablation of low-grade UTUC**. It is a viscous liquid that **gels at body temperature** and dissolves in urine over several hours, delivering mitomycin at higher concentration for longer than liquid agents.
- **Phase III single-arm trial: complete response 59% at first evaluation; 56% disease-free at 12 months**. Instilled retrograde or antegrade (antegrade had less stricture in early retrospective series).
- **Toxicities: ureteric stricture (about 44%), bone-marrow suppression, embryo-fetal toxicity.**
- **Complete endoscopic ablation remains the mainstay; UGN-101 is an alternative when tumours are endoscopically unmanageable**, after informed disclosure of toxicity.

Upper-tract BCG	UGN-101 (mitomycin gel)
High-grade, clinically non-invasive disease: CIS; high-risk favourable Ta/T1 after complete ablation	Low-grade tumours not amenable to endoscopic treatment

Concept — *Topical therapy is a contest with gravity: the upper tract drains too fast for liquid agents, so delivery (nephrostomy, catheter, reflux, or a gel that outlasts the flow) decides efficacy, while pressure and infection decide toxicity.*

Perioperative Systemic Therapy

- **Systemic strategies parallel those for bladder urothelial cancer.** Prospective trials show benefit of neoadjuvant and adjuvant therapy in high-risk disease, establishing both as **standard of care in selected patients**.
- **Rationale for neoadjuvant therapy:** baseline chronic kidney disease is common in UTUC and **worsens** after RNU, making many patients **ineligible for full-dose cisplatin postoperatively**.

PROSPECTIVE PERIOPERATIVE CHEMOTHERAPY TRIALS IN UTUC

Trial	Setting / design	Regimen by renal function	Result
EA8141	Neoadjuvant; phase II, non-randomised; high risk	Creatinine clearance (CrCl) >50: accelerated MVAC × 4; CrCl 30–49: gemcitabine–carboplatin × 4	Met end point: pathological complete response 14%; downstaging to <ypT2N0 60% ; no additional safety concern with CrCl >50
Gemcitabine + split-dose cisplatin trial	Neoadjuvant; phase II; high-grade or invasive clinical features	Estimated GFR (eGFR) ≥50 mL/min/1.73 m ²	Met end point (majority <ypT2N0; complete response in about one-fifth); most received ≥3 cycles; all proceeded to surgery 4–6 weeks later; responders had better progression-free and overall survival
POUT	Adjuvant; phase III randomised versus surveillance; after RNU, pT2–T4 N0–3 M0	CrCl >50: gemcitabine–cisplatin; CrCl 30–49: gemcitabine–carboplatin	Stopped early for efficacy. Disease-free survival HR 0.45; 3-year disease-free survival 71% vs 46%; metastasis or death HR 0.48 ; grade ≥3 adverse events 44%; 58% of cisplatin starters completed four cycles

MVAC, methotrexate, vinblastine, doxorubicin, cisplatin; HR, hazard ratio.

- **Meta-analysis of neoadjuvant therapy:** significant pathological downstaging, pathological complete response and 5-year disease-specific survival benefit; outcomes appear superior to historical RNU alone.
- **AUA/SUO: neoadjuvant platinum-based chemotherapy is a standard-of-care option for defined high-risk UTUC, especially when postoperative CrCl is expected to fall below 50.**
- **Earlier retrospective adjuvant studies showed no survival benefit**, but lacked selection criteria, standard regimens and account of renal function — biases exposed once prospective (POUT) data arrived.
- **POUT subanalyses:** lower efficacy in node-positive disease (see six cycles below) and with carboplatin → **cisplatin remains preferred**, reinforcing neoadjuvant treatment for high-risk patients with borderline renal function.

Concept — *Renal function is the hinge: RNU removes a kidney and with it cisplatin eligibility, so platinum is best given before the kidney goes; POUT proves adjuvant chemotherapy works, and its weakness with carboplatin is itself the argument for going neoadjuvant.*

Treatment of Lymph Node-Positive and Metastatic Disease

Chemotherapy

- **UTUC-specific data are limited**; randomised comparisons are not feasible because the disease is rare, so response data are extrapolated from urothelial cancer, mostly without stratification by site. In one institutional series, recurrence-free survival did not improve over two decades.
- **Postoperative renal decline in elderly patients** compromises chemotherapy after RNU — another reason for neoadjuvant treatment of high-risk tumours.
- **Regional nodal metastases → chemotherapy first as primary therapy**; withhold surgery until a good (ideally complete) radiographic response; **two further cycles after maximal response (usually six in total)**; then **consolidative surgery including lymphadenectomy** — the bladder paradigm.

CHEMOTHERAPY REGIMENS IN ADVANCED UROTHELIAL CANCER

Regimen	Position
MVAC	Highest response rate ; toxicity prevents optimal dose and duration in many; complete response rare in metastatic disease; overall survival 12–24 months
Dose-dense MVAC	All drugs given together with growth-factor (cell) support; lower toxicity , possibly better response
Newer agents (paclitaxel, ifosfamide, carboplatin, gemcitabine, vinflunine) in combination	Initial response similar to MVAC with lower toxicity; complete response rare; no head-to-head data on durability or survival
Carboplatin for cisplatin	Used for renal limitation or toxicity; results inferior
Paclitaxel–cisplatin–gemcitabine (PCG) vs gemcitabine–cisplatin (GC)	Phase III: adding paclitaxel modestly improved median overall survival; both well tolerated; benefit more pronounced in bladder primaries than upper-tract primaries
Gemcitabine–paclitaxel–doxorubicin (GTA)	Poor renal reserve ; renal function may improve after a few cycles, allowing switch to cisplatin
Cabozantinib	MET and vascular endothelial growth factor (VEGF) pathway inhibitor; encouraging early results after chemotherapy failure (Fig. 139.27); phase II ongoing

Immunotherapy

- **Mechanism**: programmed death-1 (PD-1) is an inhibitory receptor; its activation by programmed death-ligand 1 (PD-L1) inhibits T-cell proliferation and cytokine production. Blocking this axis produced remarkable activity in metastatic urothelial carcinoma, with a favourable side-effect profile and, importantly, **infrequent renal impairment**.
- **UTUC may benefit particularly: microsatellite instability (MSI) is higher in UTUC than in bladder cancer**, and high MSI is especially responsive to checkpoint inhibition.
- Before these agents, options after platinum progression were poor — a major unmet need.

CHECKPOINT INHIBITORS IN ADVANCED UROTHELIAL CANCER

Drug	Target	Approval and evidence	Bottom line
Nivolumab	PD-1	2017; phase II, after platinum	Meaningful benefit, acceptable safety; response not significantly related to PD-L1 expression

Drug	Target	Approval and evidence	Bottom line
Pembrolizumab	PD-1	2017; KEYNOTE-045 (phase III versus chemotherapy, after platinum): median overall survival 10.3 vs 7.3 months, HR 0.70	Responses durable after stopping therapy; higher PD-L1 staining not substantially better; also approved for any MSI-high solid tumour
Atezolizumab	PD-L1	First PD-L1 inhibitor active in urothelial cancer; 2016 approval after platinum, followed by first-line approval for cisplatin-ineligible disease; IMvigor211 (phase III versus chemotherapy): objective response 23% vs 22%, response duration 15.9 vs 8.3 months (not significant)	Similar response, longer duration, fewer high-grade adverse events; US urothelial indications since withdrawn
Avelumab	PD-L1 + antibody-dependent cell-mediated cytotoxicity	Platinum-refractory disease; phase Ib; pooled cohorts included UTUC patients	Unique dual action — directly lyses tumour cells
Durvalumab	PD-L1	Phase I/II, mostly pretreated	Early, durable responses independent of PD-L1; US urothelial indication since withdrawn

- Across the class, PD-L1 expression did not reliably predict benefit.

Molecular Alterations and Future of Genomic-Driven Therapy

- UTUC and bladder cancer are phenotypically similar but genomically somewhat dissimilar.
- MSI is high in UTUC, relevant now that pembrolizumab is approved for any solid tumour with high MSI.

High-grade UTUC, relative to high-grade bladder cancer

More: FGFR3 (trend), HRAS, CDKN2B mutations. **Fewer:** TP53 and RB1 mutations. **Shared:** frequent chromatin-modifier mutations

- **Comprehensive UTUC genomics: FGFR3 the most frequently mutated gene**, in both grades but more in low grade; then KMT2D, PIK3CA and TP53. Hypermutated tumours with deficient mismatch-repair proteins link to **Lynch syndrome**. APOBEC and CpG are the commonest mutational signatures. RNA sequencing defines four putative subtypes, one with high CTLA-4, PD-1 and PD-L1 expression.
- **Erdafitinib, a pan-fibroblast growth factor receptor (FGFR) inhibitor, is the first targeted agent for metastatic urothelial cancer**; phase II data in FGFR-altered tumours show meaningful response and high disease control with intermittent or continuous dosing.
- **Summary:** UTUC is chemosensitive, but standard regimens are toxic and responses not sustained; baseline chronic kidney disease, worsened by RNU, is unique to this population; checkpoint inhibitors are improving outcomes in treatment-naïve and platinum-refractory disease, and genomic-driven therapy is expanding.

Concept — UTUC is the FGFR3-rich, MSI-high, cisplatin-poor relative of bladder cancer, and each trait opens a different therapeutic door: FGFR inhibition, checkpoint blockade, and neoadjuvant timing of platinum.

Radiotherapy

- **Limited role outside palliation:** proximity of bowel to the upper tract prevents the dose needed for effective primary therapy.

FOLLOW-UP

General Principles

- **Multifocal recurrence and metastatic spread (more with dysplastic lesions) complicate follow-up.** Evaluate the bladder, the ipsilateral tract (if spared), the contralateral tract, and extra-urinary sites. The schedule depends on time since surgery, approach (kidney-sparing versus radical) and risk.
- **AUA/SUO schedules are stratified by approach and by risk (low versus high) or pathology (<pT2N0/x versus ≥pT2).**
- **Every visit:** history, examination, basic metabolic panel and **office cystoscopy** (high bladder recurrence after both conservative treatment and RNU). **After kidney-sparing: upper-tract endoscopy** for ipsilateral recurrence.
- **Urine cytology** helps detect upper-tract recurrence, especially of high-grade tumours.
- **Contrast CT or MRI of abdomen and pelvis with excretory phase** (risk of synchronous or metachronous bilateral disease; metastatic staging). After RNU, renal impairment may preclude contrast → **retrograde pyelography** if iodinated contrast or gadolinium cannot be given or the urographic phase is non-diagnostic. **Filling defects usually need ureteroscopy.** **Chest imaging** for metastatic staging.
- **Nephrology referral for diminished renal function** after RNU or other intervention; encourage smoking cessation, exercise and healthy diet.

Surveillance After Kidney Sparing and After Radical Nephroureterectomy

AUA/SUO SURVEILLANCE SCHEDULES

Group	Cystoscopy	Upper-tract endoscopy	Imaging
Kidney-sparing, low risk	Within 3 months; then at least every 6–9 months for 2 years ; then at least annually	Within 3 months; repeat at 6 months and 1 year	Upper tract at least every 6–9 months for 2 years, then annually to 5 years; then shared decision
Kidney-sparing, high risk	Within 3 months; then at least every 3–6 months for 3 years ; then at least annually	Within 3 months; repeat at least at 6 months and 1 year	Upper tract every 3–6 months for 3 years, then annually to 5 years; then shared decision
RNU, <pT2 N0 M0	Cystoscopy and cytology within 3 months; then repeated according to grade	—	CT/MRI abdomen and pelvis within 6 months; then at least annually for at least 5 years
RNU, ≥pT2 Nx/N0	Cystoscopy and cytology at 3 months; every 3–6 months for 3 years; then annually	—	Abdomen and pelvis every 3–6 months for 2 years, 6-monthly in year 3, then annually to 5 years; chest CT every 6–12 months for 5 years

Concept — *Surveillance intensity mirrors what remains at risk: kidney-sparing keeps the ipsilateral tract under endoscopic watch, high risk compresses the intervals, and after RNU the target shifts from the upper tract to the bladder and distant sites.*

KEY POINTS

- **Assess suspected UTUC with history, imaging, endoscopy, biopsy and cytology** to inform risk-stratified surgical management.
- **Lynch syndrome is present in 8–12% of newly diagnosed UTUC**; evaluate all patients by family history and tissue biomarkers (mismatch-repair immunohistochemistry) or germline testing as standard of care.

- **Predictors of advanced stage** to document: high-grade biopsy or cytology, multifocality, obstruction, tissue invasion, sessile endoscopic appearance.
- **Endoscopic management suits low-grade, low-risk tumours**, with careful technique to minimise adverse events.
- **Topical treatment for highly selected patients**: BCG in high-grade clinically non-invasive cancer; mitomycin gel in low-grade tumours not amenable to endoscopy — after counselling on serious adverse events.
- **Conservatively managed patients must understand** frequent recurrence, repeat procedures, long-term follow-up, and the progression that would change strategy.
- **RNU with bladder cuff is the most definitive treatment** for non-metastatic UTUC — for high-grade, high-risk cancers, or bulky or progressive unfavourable low-grade cancers of the proximal ureter and renal pelvis.
- **Distal ureterectomy with bladder cuff and reimplantation** for lower-ureteric tumours, particularly with high-grade or invasive features.
- **Template lymph node dissection** during extirpative surgery for high-grade UTUC.
- **Oncologic principles in open or minimally invasive surgery**: attempted R0 resection, watertight repair, no uncontrolled urinary spillage.
- **Single-dose adjuvant topical therapy** after endoscopic and extirpative surgery.
- **Neoadjuvant platinum-based chemotherapy** for all eligible high-grade, high-risk patients with borderline renal function before RNU.
- **pT2–T4 and/or N1 after surgery → refer for adjuvant treatment**: platinum chemotherapy, checkpoint inhibitors, or biomarker-directed trials.